RC CIRCUIT CHARGING: ODE AND TIME SOLUTION

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1.0 Learning Outcomes

Upon completing this practical exercise, students should be able to demonstrate the Learning Outcomes presented in Table 1. At the end of the exercise, revisit each Learning Outcome and confirm that it has been achieved.

Table 1: *Learning Outcomes for the Practical Exercise*

ID	Learning Outcome	Achieved
1	Derive the first-order ordinary differential equation (ODE) of an RC circuit from Kirchhoff's voltage law, and solve it for a step input.	☐
2	Measure the charging and discharging of the capacitor with the Arduino Uno and compare the measurements with the analytic solution.	☐
3	Solve the ODE numerically in MATLAB, and use the measured time constant to estimate the actual capacitance.	☐

2.0 Hardware Set-Up

This practical exercise uses the Arduino Uno and breadboard from the ACE-Lab Base + Sense kit, with one resistor and one capacitor. Values of R = 100 kΩ and C = 10 µF give a time constant of 1 s, which is slow enough to see clearly and fast enough for a short test. If you use an electrolytic capacitor, connect its negative lead to GND.

Connect a digital output pin (for example D7) through the resistor to the positive lead of the capacitor, and connect the other lead of the capacitor to GND. Connect the junction between the resistor and capacitor to an analogue input (for example A0). The digital output then acts as a 0 V to 5 V step, and the analogue input measures the capacitor voltage.

3.0 Practical Investigation via Simulink

MATLAB and Simulink should be used to complete this practical exercise. The ODE of the RC circuit is derived and solved by hand, the capacitor voltage is measured as it charges and discharges, and the measurements are compared with the analytic and numerical solutions. The exercise supports the ACE-CORE learning guide, ODE and Time Solution.

Derive and Solve the ODE

The same current i flows through the resistor and the capacitor. Kirchhoff's voltage law around the loop, with $i = C\,\mathrm{d}v_c/\mathrm{d}t$, gives the first-order ODE:

$$RC\frac{dv_c}{dt} + v_c = v_{in}$$

For a step from 0 V to V = 5 V at t = 0, with the capacitor initially discharged, the solution is:

$$v_c(t) = V(1 - e^{-t/RC})$$

When the input is then switched back to 0 V with the capacitor charged to V_0, the capacitor discharges as:

$$v_c(t) = V_0\, e^{-t/RC}$$

1. Derive the ODE above from Kirchhoff's voltage law, showing each step.
2. Solve it by hand for the charging step, by finding the homogeneous and particular solutions and applying the initial condition.
3. Calculate the time constant RC for your components, and the capacitor voltage you expect at $t = RC$, $3RC$ and $5RC$.

Measure the Charging and Discharging

[Screenshot to insert] *Pulse Generator (period 12 s, 50% duty) → Digital Output (D7); Analog Input (A0) → Gain 5/1023 → Scope logging the capacitor voltage v and time t.*

Figure 1: *Simulink Model for the RC Circuit Test*

1. Build the model in Figure 1. A Pulse Generator with a period of 12 s and a pulse width of 50% drives the digital output, giving 6 s high and 6 s low. Convert the analogue reading to volts with a Gain of 5/1023.
2. Set the sample time to 0.01 s and log the input, the capacitor voltage and the time to the workspace as u, v and t.
3. Make sure the capacitor is discharged by connecting both leads to GND briefly. Then run the model for 12 s.

The theoretical response for R = 100 kΩ and C = 10 µF is shown in Figure 2, quantised to the 10-bit ADC. Your measurement should have the same shape, but the time constant may differ by up to about 20%, because electrolytic capacitors commonly have a tolerance of ±20%.

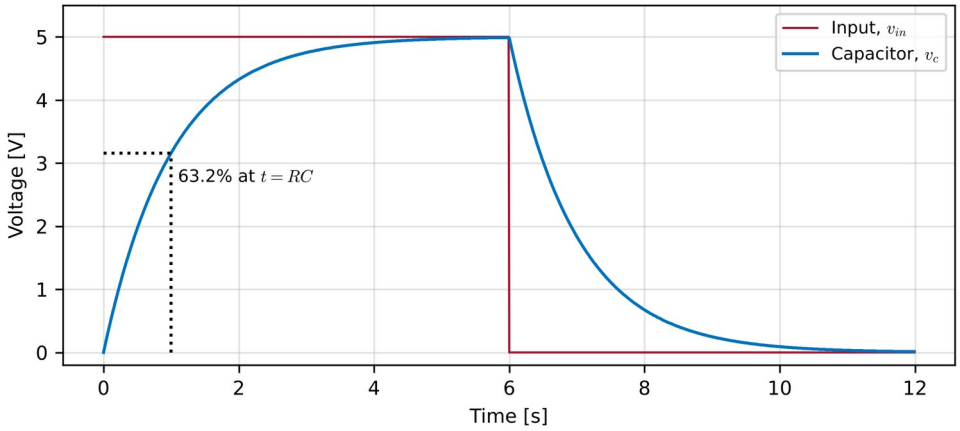


Figure 2: *Theoretical Charging and Discharging of the Capacitor for RC = 1 s*

Compare the Measurement with the Solutions

MATLAB Script: comparing the measured, analytic and numerical solutions

```
v_k = v.signals.values(:);  t_k = t.signals.values(:);
R = 100e3;  C = 10e-6;  RC = R*C;  V = 5;       % nominal values

%% Analytic solution (charging, first 6 s)
ch = t_k < 6;
v_an = V*(1 - exp(-t_k(ch)/RC));

%% Numerical solution of the ODE with ode45
odefun = @(tt, vc) (V - vc)/RC;
[t_num, v_num] = ode45(odefun, [0 6], 0);

%% Time constant from the measurement
RC_meas = t_k(find(v_k >= 0.632*V, 1))          % measured time constant [s]
C_est   = RC_meas/R                             % estimated capacitance [F]

plot(t_k(ch), v_k(ch), 'k.', t_k(ch), v_an, 'b', t_num, v_num, 'r--');
grid on; xlabel('Time [s]'); ylabel('Capacitor voltage [V]');
legend('Measured', 'Analytic (nominal RC)', 'ode45', 'Location', 'southeast');
```

Table 2: *Results for Your RC Circuit*

Quantity	Expected (nominal)	Measured
Voltage at $t = RC$ (V)	3.16	
Voltage at $t = 3RC$ (V)	4.75	
Time constant from charging (s)	1.00	
Time constant from discharging (s)	1.00	
Estimated capacitance (µF)	10	

1. Use the measured time constant to estimate the actual capacitance, and compare it with the marked value and its tolerance.

2. Derive the solution for a capacitor that starts at 2 V, and check it by charging the capacitor partly before starting the test.
3. The analogue input of the Arduino Uno has an input resistance of about 100 MΩ. Explain why it hardly affects the measurement.

4.0 Concluding Remarks

This practical exercise modelled an RC circuit with a first-order ODE, solved it analytically and numerically, and compared the solutions with measurements. Using MATLAB, Simulink and ACE-Lab hardware, the capacitor voltage was measured as it charged and discharged, and the time constant was read from the 63.2% point.

The analytic solution, the numerical solution and the measurement agreed in shape, with differences in the time constant explained by component tolerances. The same first-order ODE describes many other systems, including the speed of the DC motor, so the method used here applies directly to the motor models used in later exercises.

Recommended Citation

If you have used this practical exercise in your teaching, learning or research, please cite it using the reference below, adapting the format as required by your preferred referencing style.

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